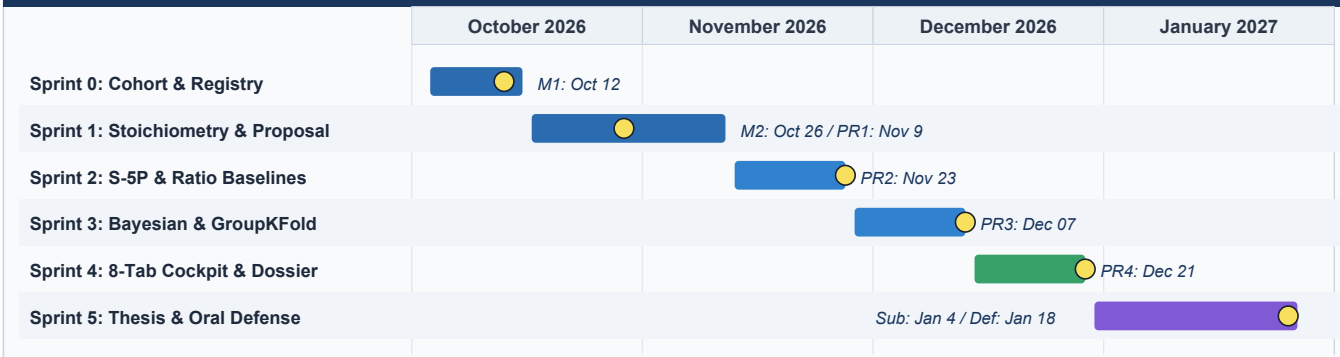
VeriCBAM: Capstone Project Work Plan, Risk Matrix & Execution Roadmap

Multimodal Evidence Fusion & Decision Support for CBAM Emissions Consistency Assessment

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VeriCBAM Execution Gantt Chart & Milestone Timeline (Winter Semester 2026/2027)



1. Project Execution Schedule & Academic Milestones (Winter 2026/2027)

Phase / Sprint	Timeline	Key Technical & Academic Activities	Primary Deliverable / Milestone
Milestone 1 & Sprint 0	Oct 03 – Oct 14, 2026	Topic alignment, EEA E-PRTR v16 cohort extraction (30 facilities, 177 plant-years), GEM capacity & boundary registry curation.	M1: Topic Approval (Oct 12, 2026) & Cleaned Layer A Registry
Milestone 2 & Sprint 1	Oct 15 – Nov 09, 2026	Proposal formalization; chemical stoichiometry engines (`cement_models.py`, `steel_models.py`), boundary gas-export rules, XML/XLSX declaration parser.	M2: Proposal Approval (Oct 26, 2026) & Progress Report 1 (Nov 09)
Sprint 2	Nov 10 – Nov 23, 2026	Copernicus Sentinel-5P TROPOMI Level-2 NO_2 extraction (`copernicus_client.py`), cloud-fraction gating, leave-one-year-out NO_x/CO_2 ratio & pooled-MAD baselines.	M4: Progress Report 2 (Nov 23, 2026) & 5-Stream Feature Pipeline
Sprint 3	Nov 24 – Dec 07, 2026	Weighted Bayesian log-odds fusion (Model E1), confidence gating ($C_{overall} < 0.50$), 5-fold `GroupKFold` logistic triage (Model E2), 325-case graded benchmark & bootstrap CIs.	M5: Progress Report 3 (Dec 07, 2026) & Empirical Ablation Suite
Sprint 4	Dec 08 – Dec 21, 2026	8-tab Streamlit Decision Cockpit (`src/ui/app.py`), standalone zero-history supplier pre-check, 1-click audit dossier exporter, `pytest` verification suite.	M6: Progress Report 4 (Dec 21, 2026) & Live Cloud Cockpit
Sprint 5 & Defense	Dec 22 – Jan 18, 2027	Master thesis manuscript finalization, end-to-end reproducibility verification (`reproduce_all.py`), and oral examination slides.	M7: Final Submission (Jan 04, 2027) & M8: Defense (Jan 18, 2027)

2. Technical Risk Management Matrix

Identified Risk	Severity	Likelihood	Impact on Project	Proactive Mitigation Strategy
Persistent Cloud Obscuration in Sentinel-5P	High	High	Overpass data gaps over North/Central Europe during winter months.	Use `qa_value >= 0.50` with explicit cloud fraction confidence weighting ($C_{sat} = 1 - f_{cloud}$) and an `Insufficient Evidence` abstention gate ($C_{overall} < 0.50$).
Transferred Process Gases in Steelworks	High	High	False physical violation flags on BF-BOF plants exporting blast furnace gas (BFG) to off-site power plants.	Map legal operating boundaries from national IED permits; set `stoich_applicable = False` ($\hat{p}$=NaN) and evaluate via co-pollutant and historical streams.
Biomass Emission Reporting Missingness	Medium	High	Inability to isolate biogenic CO_2 in public registry records.	Standardize on total direct CO_2 as the primary regulatory proxy across the 2018–2023 historical panel series.
Facility Data Leakage in ML Models	Critical	Low	Artificially inflated cross-validation benchmark metrics.	Enforce strict 5-fold `GroupKFold` grouped by `facility_id`, isolating all multi-year records of any given plant between training and test folds.
Construct Validity & Proxy Mismatch	High	Medium	Facility annual mass (tCO_2) does not directly equal product specific intensity (tCO_2e/t).	Explicitly document proxy boundaries; build a standalone supplier pre-screening module that accepts declared physical production volume (Q).
Synthetic Benchmark Circularity	Medium	Medium	Rule-based perturbations may not capture all real-world evasion patterns.	Evaluate across five graded understatement magnitudes (δ in {5%, 10%, 20%, 30%, 50%}) plus cross-route misclassifications, reporting Recall at fixed FPR budgets (5%, 10%).

3. Threats to Validity & Methodological Boundaries

1. Construct Validity (Facility Mass vs. Product Intensity): E-PRTR reports annual absolute facility emissions (tCO_2), whereas CBAM regulates specific embedded emissions per tonne of product (tCO_2e/t). Without declared production output Q, a longitudinal mass drop cannot distinguish between an emissions understatement and a genuine economic reduction in factory utilization.
2. Confidentiality of Third-Country CBAM Declarations: Because real non-EU transitional CBAM declarations are protected as confidential business information by the European Commission, empirical testing relies on an authentic European industrial reference cohort combined with controlled graded perturbations.
3. Cohort Depth vs. Breadth: The reference panel focuses on 30 major European installations (15 Cement, 15 Steel) across 10 EU Member States, prioritizing verified process-route and capacity metadata depth over unverified bulk scraping.

4. Core Software Deliverables & Resource Management Plan

• Software & Academic Deliverables:

1. VeriCBAM Core Engine (`src/`): Modular Python package implementing thermodynamic mass-balance bounds (`cement_models.py`, `steel_models.py`), leave-one-year-out co-pollutant baselines (`nox_baseline.py`), Copernicus Sentinel-5P extraction (`copernicus_client.py`), and Bayesian log-odds fusion (`evidence_fusion.py`).
2. 8-Tab Interactive Streamlit Decision Cockpit (`src/ui/app.py`): Covering (1) System Overview & Architecture, (2) Facility Cohort & Geospatial Inspector, (3) Stoichiometric Feasibility Sandbox, (4) Standalone Supplier Plausibility Pre-Check, (5) Satellite Inspector, (6) Bayesian Evidence Fusion & Audit Dossier, (7) Empirical Benchmark Evaluation & Ablation, and (8) Capstone Proposal, References & Roadmap.
3. Reproducibility & Verification Suite: Single-command reproduction script (`scripts/reproduce_all.py`) and automated `pytest` unit/integration suite (`tests/`).

• Resource & Governance Plan: Executed on a standard Python 3.12 64-bit workstation without specialized GPU requirements, utilizing public EEA E-PRTR v16 data, Copernicus Data Space Ecosystem (CDSE) APIs, and Global Energy Monitor (GEM) trackers under bi-weekly supervision sprints with Dr. Humera Noor.

5. Technical Extension Roadmap: Multi-Sensor Earth Observation (Stage 2 Horizon)

To overcome coarse trace-gas resolution (5.5 × 3.5 km) and eliminate reliance on self-reported production volume (Q) for third-country suppliers lacking historical baselines, the project roadmap defines a Stage 2 Multi-Sensor EO extension:

1. Thermal Infrared (TIR) Operating Timelines (USGS Landsat-8/9 TIRS 30 m & NASA FIRMS VIIRS 375 m): Detects persistent radiative heat anomalies over cement rotary kilns (1,400°C core / 200–350°C shell) and steel blast furnaces (1,500°C hot metal), reconstructing an independent monthly operating calendar.
2. High-Resolution Optical Freight & Stockpile Logistics (PlanetScope / VHR Optical): Quantifies rail-yard freight activity and raw-material yard contrast to independently verify physical plant throughput (Q).
3. All-Weather Synthetic Aperture Radar (Copernicus Sentinel-1 C-Band SAR): Penetrates winter cloud cover via dual-polarization metallic double-bounce backscatter over storage yards and gantry cranes.